# UNIVERSITY OF THE PUNJAB

**B.S. 4 Years P... Sixth Semester Spring 2025** Roll No. [redacted]

**Paper: Quantum Mechanics-I** **Course Code: PHYS-3301** **Time: 3 Hrs. Marks: 6**

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Write short answer of the following questions: (15x2=30)**

1. What is a Hilbert space, and why is it important in Quantum Mechanics?
2. Define Hermitian operators. How do they relate to the measurement in quantum mechanics?
3. What is the difference between unitary operators and Hermitian operators?
4. Explain the representation of quantum states in a discrete basis.
5. Describe the position representation and momentum representation in quantum mechanics.
6. State and explain the postulates of Quantum Mechanics.
7. What is the generalized uncertainty principle? Explain its significance.
8. Discuss the importance of Dirac notation in quantum mechanics.
9. What is the physical meaning of a Hermitian operator in quantum mechanics?
10. Derive the solution to the Schrödinger equation for a free particle in one dimension.
11. Explain the concept of a potential step and its solution in quantum mechanics.
12. What are Hermite polynomials, and how are they related to the solution of the Simple Harmonic Oscillator?
13. Describe the angular momentum operator in quantum mechanics. What are its key properties?
14. What is the energy eigenvalue of a simple harmonic oscillator? Also write the Hamiltonian of the simple harmonic oscillator
15. State and explain the significance of the Laguerre polynomials in the solution of the Hydrogen atom.

**Q.2. Solve the following. (3x10=30)**

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1. Derive the solution of the Schrödinger wave equation for a free particle in one dimension. Discuss the significance of the wave function, its boundary conditions, and the normalizability of the wave function. How do the energy eigenvalues arise, and what do they represent physically?
2. Derive the quantum mechanical treatment of angular momentum. Explain how the angular momentum operators $L^2$ and $Lz$ work in spherical coordinates. Discuss their eigenvalues and eigenfunctions in the context of spherical harmonics and explain the implications for the quantum state of a particle.
3. Solve the Schrödinger equation for the Hydrogen atom. Derive the energy eigenvalues and eigenfunctions of the system. Discuss the quantum numbers that arise from the solution, and explain how they relate to the physical properties of the atom, including the use of Laguerre polynomials in the radial wave function.

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# UNIVERSITY OF THE PUNJAB

**BS (After Associate Degree) : Sixth Semester – Fall 2023**

Roll No. [redacted]

**Paper: Quantum Mechanics-I** (Common with BS4Y 6th)

**Course Code: PHYS-3301** Time: 3 Hrs. Marks: 60

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Write short answer of the following questions:** (10x3=30)

(i)- What is meant by operators and state vectors in quantum mechanics? Also explain dimensionality and basis.

(ii)- What is meant by discrete and continuous spectrum? Explain it by considering the example of a general potential.

(iii)- State any three postulates of quantum mechanics.

(iv)- If $x$ and $p$ are position and momentum variables and

$$\hat{A} = \frac{1}{\sqrt{2m\hbar\omega}}(m\omega\hat{x} + i\hat{p})$$

Show that $[\hat{A}, \hat{A}^{\dagger}] = 1$.

(v)- Find out classical amplitude for a particle undergoing simple harmonic motion if its energy is $\hbar\omega/2$.

(vi)- A particle is in state given by wave function,

$$\psi(x,t) = e^{-ax^2-bt}$$

where a and b are constants.
Is it a stationary state? Find expectation values of position and momentum operators.

(vii)- Show that if an operator $\hat{A}$ has an eigen value $a \neq a^*$, then $\hat{A}$ is not Hermitian.

(viii)- Given that in spherical polar coordinates $\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$, show that

$$[\hat{L}_z, \sin\phi] = -i\hbar\cos\phi$$

(ix)- The operator for z-component of angular momentum is $\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$. Determine whether or not $\sin m\phi$ is its eigen function?

(x)- State results of Stern-Gerlach experiment.

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**Q.2. Solve the following.** (3x10=30)

(i)-State and prove generalized uncertainty relation (Uncertainty principle for operators).

(ii)-Differential equation for simple harmonic oscillator obtained after solving Schrodinger wave equation is given by,

$$H''(z) - 2zH'(z) + (\lambda - 1)H(z) = 0$$

Solve this equation to find energy eigen values and wave functions of simple harmonic oscillator.

(iii)-Discuss the significance of ladder operators $\hat{L}_{\pm}$. Show that they have following property.

$$\hat{L}_+\psi(l,m) = \sqrt{l(l+1) - m(m+1)}\,\hbar\psi(l,m+1)$$

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# UNIVERSITY OF THE PUNJAB

**B.S. 4 Years Program / Sixth Semester – Spring 2024**

Roll No. 49..........

Paper: Quantum Mechanics-I
Course Code: PHYS-3301

(RE-CONDUCT)

Time: 3 Hrs. Marks: 60

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Write short answer of the following questions: (15x2=30)**

I. Prove that for a Hermitian operator, the eigen vectors corresponding to different eigen values are orthogonal.

II. Write the expression for Identity operator (completeness relation) in Dirac notation.

III. Prove the following properties of Dirac notation. i) Schwarz inequality ii) Triangle inequality.

IV. What is the condition for which the sum of two projection operators $\widehat{P_1}$ and $\widehat{P_2}$ is also a projection operator?

V. What is Correspondence Principle.

VI. Write down the expression for momentum operator in the position representation.

VII. Prove that $[L_x, L^2] = 0$

VIII. Can we determine z-component of angular momentum and azimuthal angle with precision simultaneously?

IX. Write any two postulates of Quantum Mechanics.

X. Prove that i) Expectation value of a Hermitian operator is always real
ii) Expectation value of an anti-Hermitian operator is imaginary.

XI. *What is the difference between the energy spectra of a bounded system and an unbounded system?*

XII. Write down the expression for time rate of change of expectation value of an operator $A$

XIII. Write down the statement for principle of superposition in Quantum mechanics.

XIV. Write down the equation of continuity showing the conservation of probability in Quantum mechanics.

XV. Define Unitary operator.

**Solve the following.**

Q.2. Develop the relation for generalized uncertainty principle for operators $\hat{A}$, $\hat{B}$ and using $\hat{A}$ and $\hat{B}$ as position and momentum operators in the generalized relation prove the Heisenberg Uncertainty principle. (8+2)

Q.3. *Find the eigen function and corresponding eigen values for a 1-D harmonic oscillator using the algebraic method and prove that energy spectrum is both discrete and degenerate.* (10)

Q.4. What is a Central potential? Evaluate Radial wave equation for Hydrogen atom. (10)

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# UNIVERSITY OF THE PUNJAB

**B.S. 4 Years Program / Sixth Semester – Spring 2023**

**Paper: Quantum Mechanics-I** **Course Code: PHYS-3301** **Time: 3 Hrs. Marks: 60**

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Answer the following short questions.** **(10x3=30)**

i. Prove that eigen value of Hermitian operator is real.

ii. Define wave function. What are properties of a wave function?

iii. Find out if the function $\Psi(x) = e^{ax}$ is an eigen function of x-component of momentum operator.

iv. Consider two states

$$|\Psi_1> = 2i\,|\phi_1> + |\phi_2> \; -\alpha\,|\phi_3> + 4|\phi_4>$$
$$|\Psi_2> = 3\,|\phi_1> - i|\phi_2> \; +5\,|\phi_3> - |\phi_4>$$

Where $|\phi_1>, |\phi_2>, |\phi_3>$ and $|\phi_4>$ are orthonormal kets and is $\alpha$ constant. Find the value of $\alpha$, if $|\Psi_1>$ and $|\Psi_2>$ are orthogonal.

v. What is physical significance of Hamiltonian in Quantum Mechanics?

vi. Define complementary and correspondence principles.

vii. Write down physical significance of ladder operators in Quantum mechanics.

viii. In case of orbital momentum operator, can $\hat{L}^2$ and $\hat{L}_z$ both be simultaneously measured? Explain the reason.

ix. Compute $[\hat{S}_z, \hat{S}_x]$=?

x. Prove That $\hat{a}|n> = \sqrt{n}\,|n-1>$. Where $\hat{a}$ is annihilation operator.

**Answer the following questions.**

**Q. 2.** (a) Explain the Stern-Gerlach Experiment in detail. (6)

(b) Given that in spherical coordinates $\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$, determine whether or not $\sin m\phi$ and $e^{im\phi}$ are eigen functions of $\hat{L}_z$. (4)

**Q. 3.** Find the reflection and transmission coefficients R and T for 1 dimensional potential step for the case when E > $V_o$, where

$$V(x) = \begin{cases} 0, & for\ x < 0 \\ V_o, & for\ x \geq 0 \end{cases}$$

And also show that R + T = 1. (3+3+4)

**Q. 4.** Define the term central potential. Starting with the time independent Schrodinger's wave equation, obtain an expression of radial wave function. (2+8)

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**UNIVERSITY OF THE PUNJAB**

**BS (After Associate Degree) : Sixth Semester – Spring 2023**

Roll No. ..................

**Paper: Quantum Mechanics-I** **Course Code: PHYS-3301** **Time: 3 Hrs.** **Marks: 60**

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Write short answer of the following questions:** **(10x3=30)**

**(i)**- State any three postulates of quantum mechanics.

**(ii)**- Define Hilbert space and hence Fock space in terms of Hilbert space. State similarities and differences in both spaces.

**(iii)**- What is meant by square integrable functions? Explain ket and bra notations (Dirac notations) briefly.

**(iv)**- What is importance of simple harmonic oscillator? Give it classical theory.

**(v)**- Write down Hamiltonian of simple harmonic oscillator. Define new operators,

$$\hat{P} = \frac{\hat{p}}{\sqrt{m\hbar\omega}} \quad \text{and} \quad \hat{Q} = \hat{x}\sqrt{\frac{m\omega}{\hbar}}$$

Show that Hamiltonian can be expressed as,

$$\hat{H} = \frac{\hbar\omega}{2}(\hat{Q}^2 + \hat{P}^2)$$

**(vi)**- A simple harmonic oscillator in its ground state has wave function,

$$\psi_0(x) = \frac{\sqrt{a}}{\pi^{1/4}} e^{-a^2x^2/2}, a^4 = \frac{4\pi^2 mk}{h^2}$$

Given that $\xi = ax$, $\int_0^\infty e^{-\xi^2} d\xi = \frac{\sqrt{\pi}}{2}$ and $\int_0^1 e^{-\xi^2} d\xi = (0.84)\frac{\sqrt{\pi}}{2}$

Show that probability of finding the particle outside the classical region is 16%.

**(vii)**- Consider the normalized wave function,

$$\psi = \sqrt{\frac{a}{\pi}} e^{-\frac{a}{2}(x-x_0)^2}$$

Find expectation value $\langle \hat{x} \rangle$.

**(viii)**- Given that in spherical polar coordinates $\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}$, show that

$$[\hat{L}_z, \cos\phi] = i\hbar\sin\phi$$

**(ix)**- Define ladder operators of angular momentum. Evaluate $[\hat{L}_z, \hat{L}_+]$.

**(x)**- What is principle of Stern-Gerlach experiment.

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**Q.2. Solve the following.** **(3x10=30)**

**(i)**-Given two operators $\hat{A}$ and $\hat{B}$ such that $[\hat{A}, \hat{B}] = i\hat{C}$, then prove that the uncertainties $\Delta\hat{A}$ and $\Delta\hat{B}$ in an arbitrary state are related by,

$$\Delta\hat{A}\Delta\hat{B} \geq \frac{1}{2}|\langle\hat{C}\rangle|$$

**(ii)**-Solve the Schrodinger wave equation for simple harmonic oscillator and show that it reduces to Hermite differential equation.

**(iii)**-What is relation of spherical harmonics with orbital angular momentum? Find eigen functions of angular momentum operator.

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# UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Sixth Semester – Spring 2022

Roll

Paper: Quantum Mechanics-I    Course Code: PHYS-3301    Time: 3 Hrs.    Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (10x3=30)

i. What is physical significance of a wave function? Some wave functions are given below:

$$\Psi_1(x) = x^2 , \Psi_2(x) = e^{ix}$$

Are these functions physically acceptable or not?

ii. Show that $\Psi(x) = e^{-\frac{x^2}{2}}$ is eigen function of $(\frac{\partial^2}{\partial x^2} - x^2)$ and find eigen value.

iii. Check whether momentum operator $\hat{p}_x$ and position operator $\hat{x}$ commute? What conclusion you draw from result?

iv. What is physical significance of Hamiltonian in Quantum Mechanics?

v. Define complementary and correspondence principles.

vi. Write down any three postulates of Quantum Mechanics.

vii. Derive the equation of continuity for Schrodinger wave equation.

viii. Given that in spherical coordinates $\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$, show that $[\hat{L}_z, \cos\phi] = i\hbar \sin\phi$

ix. Name the three pairs of canonically conjugate quantities.

x. Prove That $\hat{H}\, \hat{a}^\dagger|n\rangle = (E_n + \hbar\omega)\, \hat{a}^\dagger|n\rangle$. Where $\hat{a}^\dagger$ is the creation operator.

Solve the following.

Q. 2. Consider a system whose quantum mechanical wave function is given by

$$\Psi = \frac{\sqrt{3}}{3}\psi_1 + \frac{2}{3}\psi_2 + \frac{\sqrt{2}}{3}\psi_3$$

Where $\psi_1, \psi_2, \psi_3$ are all orthonormal wave functions

a) Show that wave function is normalized. (2)

b) Calculate the probability of finding the system is each of the states $\psi_1, \psi_2, \psi_3$. (6)

c) What is value of total probability? (2)

Q.3. Solve Schrodinger wave equation for a particle in one dimensional box. Also find energy eigen values and normalized wave function. (3+3+4)

Q.4. (a) State and prove Ehrenfest's Theorem. (6)

(b) Prove that eigen value of Hermitian operator is real. (4)

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# UNIVERSITY OF THE PUNJAB

**B.S. 4 Years Program : Fifth Semester – Fall 2021**

Roll No. ................

Paper: Quantum Mechanics-I    Course Code: PHY-305    Time: 3 Hrs.    Marks: 60

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**Q.1. Give short answers to the following questions.** (10x3=30)

*i.* How do you define term "Wave Function"in quantum mechanics. Write down its physical significance.

*ii.* Show that Hamiltonian operator and momentum operator commutes with each other.

*iii.* Explain any three postulates of quantum mechanics.

*iv.* Show that is an eigen function of an operator and determine eigen value if

*v.* Write down physical significance of position momentum uncertainty.

*vi.* What is zero-point energy? Calculate zero-point energy of one-dimensional harmonic oscillator.

*vii.* Consider two states

Find the value of 'a' so that and are orthogonal.

*viii.* State correspondence principle.

*ix.* Show that

*x.* Define ladder operator in quantum mechanics. Give their importance in any physical phenomenon.

**Answer the following questions in detail.**

Q2. (a). Derive equation of continuity and explain its physical significance in hydrodynamics system.

(b). If is normalized wave function. Find the expectation value of where $0<x\leq10$.
(5+5)

Q3. A particle of mass m and total energy $E>V_0$ strikes a potential step from left where
Calculate reflection and transmission coefficient. Also show that R+T=1 (10)

Q4. (a) Obtain eigen value of .

(b) Can z-component of angular momentum and azimuthal angle can be measured with precision at the same time. (6+4 )

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# UNIVERSITY OF THE PUNJAB

**B.S. 4 Years Program : Fifth Semester – 2020**

Roll N

Paper: Quantum Mechanics-I
Course Code: PHY-305 Part – II
Time: 2 Hrs. 45 Min. Marks: 50

**ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED**

**Q.2. Give short answers of the following:** (5x4=20)

*i)* Differentiate between observable and operator?

*ii)* Write the properties of an adjoint operator.

*iii)* Define complementarity and correspondence principles.

*iv)* Define a) uncertainty relation b) expectation value.

*v)* Define quantum mechanical tunneling.

**Answers the following questions.** (3x10=30)

**Question No. 3**

A beam of electrons having energy E is incident on a step potential of height $V_0$. Find reflection and transmission coefficients when $E > V_0$. Show that $R + T = 1$. (10)

**Question No. 4**

a) *Solve schrodinger's wave equation for a particle in one dimensional box.* (5)

b) Find energy eigen values and normalized wave function. (5)

**Question No. 5**

a) What is angular momentum. Define operator for angular momentum. Also find commutation relation for y and z components of angular momentum. (5)

b) Write down postulates of Quantum Mechanics. (5)

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# UNIVERSITY OF THE PUNJAB

**Fifth Semester – 2019**

**Examination: B.S. 4 Years Program**

**Roll No. ..................**

**PAPER: Quantum Mechanics-I**
**Course Code: PHY-305 Part – II**

**MAX. TIME: 2 Hrs. 45 Min.**
**MAX. MARKS: 50**

**ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED**

**Q2:** Give short answers to the following questions: **(4x5 = 20)**

i. Define a) expectation value b) eigen value of an operator

ii. What is wave mechanics?

iii. Explain degenerate spectrum.

iv. What is importance of angular momentum in quantum mechanics?

v. What is essence of correspondence principle?

**Q3:** (a) Show that eigen functions corresponding to different eigen values are orthogonal.

(b) Define Skew operator and hermitian operator. **(7+3)**

**Q4:** (a) Derive time independent Schrodinger's wave equation for a single particle.

(b) Write down three basic postulates of Quantum Mechanics. **(7+3)**

**Q5:** Find reflection coefficient and transmission coefficient of a beam of particles with energy greater than height of step potential. **(10)**

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# UNIVERSITY OF THE PUNJAB

**Fifth Semester 2018**
**Examination: B.S. 4 Years Programme**

Roll No. 0142.21........

**PAPER: Quantum Mechanics-I**
**Course Code: PHY-305**

**TIME ALLOWED: 2 hrs. & 30 mins.**
**MAX. MARKS: 50**

***Attempt this Paper on Separate Answer Sheet provided.***

**SUBJECTIVE TYPE**

i.

**Q2:** Give short answers to the following questions: **(4x5 = 20)**

i. What is zero point energy, If a classical oscillator has energy $\frac{1}{2}\hbar\omega$, What is its amplitude?

ii. Define degenerate eigen values, non-degenerate eigen values, linear dependent functions and linear independent functions.

iii. Describe Correspondence principle.

iv. State Hilbert space and give two of its examples.

v. Write physical significance of Uncertainty principle.

**Q3:** Define the term Central potential. Starting with the time independent Schrodinger's wave equation, obtain an expression of radial wave function. **(10)**

**Q4:** (a) If two operators have simultaneous eigen function, then these operators commute

(b) Write down three postulates of Quantum Mechanics. **(7+3)**

**Q5:** (a) Find eigen value and eigen function of z-component of angular momentum.

(b) Prove that $[\hat{L}_z, \sin\phi] = -i\hbar\cos\phi$ **(7+3)**

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# UNIVERSITY OF THE PUNJAB

Fifth Semester 2017
Examination: B.S. 4 Years Programme

Roll No. ...

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

*Attempt this Paper on Separate Answer Sheet provided.*

**SUBJECTIVE TYPE**

Q.2 Give short answers to the following questions.(4x5=20)

i. Define Hermitian operator and prove that $\hat{x}$ is hermitian.

ii. Show that the Eigen value spectrum of $\hat{L}_x$ is $m\hbar$. Where m has integer values.

iii. Describe the terms briefly, complete set of linear independent objects, degenerate Eigen values.

iv. Briefly explain the physical interpretation of ladder operator in Quantum mechanics.

v. What is a zero point energy? If a classical oscillator has energy $\frac{1}{2}\hbar\omega$, what is its amplitude?

Q.3 (a) State and prove uncertainty principle in Quantum mechanics. 2.26

(b) Show that $(\Delta A)^2$ is always real number.(6+4)

Q.4 (a) Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in Cartesian coordinates and convert them into spherical polar coordinates.

(b) Also calculate eigen value spectrum of $\hat{L}_z$ .(6+4)

Q.5 (a) State and prove Ehrenfest's Theorem.(2+5,3) 4.4

(b) Write down three canonical congugate observable.

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# UNIVERSITY OF THE PUNJAB

Fifth Semester 2016

Examination: B.S. 4 Years Programme

Roll No. 009583

**PAPER: Quantum Mechanics-I**

**Course Code: PHY-305**

**TIME ALLOWED: 2 hrs. & 30 mins.**

**MAX. MARKS: 50**

***Attempt this Paper on Separate Answer Sheet provided.***

## SUBJECTIVE TYPE

**Q.2.** Write short answers of the following questions (4x5=20)

1. Write down the statement of any two postulates of quantum mechanics.
2. Write down the expression for the operators for angular momentum.
3. What is expectation value? Write down its expression.
4. Discuss the superposition principle.
5. Define Hermitian operator. Give an example.

**Q.3. Explain briefly** (3x10=30)

1. Derive equation of continuity in quantum mechanica. (10)
2. State and prove uncertainty principle in quantum mechanics. (10)
3. **A particle enters a potential well such that**

**$V=0$ when $-\infty<x<0$ and $V=V_0$ for $0<x<a$ and $V=0$ for $a<x<\infty$. Solve time independent Schrödinger wave equation and show that R+T=1.** (10)


−a < x ≤ a

$V_0 = 0$


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5th

# UNIVERSITY OF THE PUNJAB

Fifth Semester 2014
Examination: B.S. 4 Years Programme

Roll No. ....................

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

PAPER: Quantum Mechanics-I
Course Code: PHY-305

Attempt this Paper on Separate Answer Sheet provided.

## SUBJECTIVE TYPE

Q.2 Give short answers to the following questions. (4x5=20)

i. Explain the term "Zero point" energy of simple harmonic oscillator. — 3.22

ii. Briefly explain the significance of Ladder operators in Quantum mechanics.

3.31 — iii. State and explain principle of superposition in quantum mechanics.

iv. If a classical oscillator has energy $\frac{1}{2}\hbar w$, what is its amplitude?

v. Using generalised uncertainty principle $\Delta A \Delta B \geq \frac{1}{2}|\langle[\hat{A},\hat{B}]\rangle|$ calculate uncertainty relation between components of position x and $p_x$

2.16 Q.3 Prove that the eigenfunctions of a hermitian operator belonging to distinct eigenvalues are orthogonal. (10)

Q.4 Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in Cartesian coordinates and convert them in spherical polar coordinates. Also calculate the eigen value spectrum of $\hat{L}_z$. (6+4)

Q.5 Let $\hat{A}$ and $\hat{B}$ be two operators and let $\Delta A$ and $\Delta B$ be uncertainties in the measurement of $\hat{A}$ and $\hat{B}$ respectively. Show that $(\Delta A)^2(\Delta B)^2 \geq |\frac{1}{2i}\langle[\hat{A},\hat{B}]\rangle|^2$ (10)

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# UNIVERSITY OF THE PUNJAB

Fifth Semester 2013
Examination: B.S. 4 Years Programme

Roll No. ....2681........

PAPER: Quantum Mechanics-I
Course Code: PHY-305

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

**Attempt this Paper on Separate Answer Sheet provided.**

## SECTION II (Subjective)

2. Give short answers to the following questions

(4×5=20)

i. Show that $(\Delta A)^2$ is always non negative real number
ii. Briefly explain the significance of Ladder operators in Quantum mechanics
iii. What is meant by Degenerate eigen values? Explain with the help of an example
iv. If a classical oscillator has energy $\frac{1}{2}\hbar\omega$, what is its amplitude?
v. Define the terms Hermitian operator, Wave front, Plane wave, Wave packet.

3. What is meant by Eigen value spectrum? Find out the eigen value spectrum of Hamiltonian $\hat{H}$ for simple harmonic oscillator.

(2+8)

4. Write down the expressions for the angular momentum operators $\hat{L}_x, \hat{L}_y, \hat{L}_z$ in cartesian coordinates and convert them in spherical polar corrdinates. Also calculate the eigen value spectrum of $\hat{L}_z$.

(6+4)

Repeat 5. Define the term central potential. Starting with the time independent Schrodinger wave equation, obtain the Radial wave equation for central potentials.

(10)

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# UNIVERSITY OF THE PUNJAB

BS (After Associate Degree): Sixth Semester – Spring 2025

Roll No. ..................

Paper: Quantum Mechanics-I    Course Code: PHYS-3301    Time: 3 Hrs.    Marks: 60

**THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED**

**Q.1. Write short answer of the following questions: (15x2=30)**

(i)- What are mathematical requirements that a wave function must satisfy to represent a physical system?

(ii)- What is Gaussian wave packet? Write the corresponding wave function and normalize it.

(iii)- What are dimensions of Hilbert space?

(iv)- What is analogy between quantum mechanical commutation bracket and classical Poisson Bracket?

(v)- Define transformation matrix. What is unitary transformation?

(vi)- Prove that for Dirac delta function $x\delta(x) = 0$.

(vii)- Show that the matrices representing operator for $\hat{x}$ and $\hat{p}$ in N-space for a harmonic oscillator obey usual commutation relation $[\hat{x},\hat{p}] = i\hbar$.

(viii)- What is importance of angular momentum in quantum physics?

(ix)- Verify that spherical harmonics $Y_{lm}(\theta,\varphi)$ are eigen states of parity operator with eigen value $(-1)^l$.

(x)- The orbital momentum or spin-angular momentum separately does not commute with Hamiltonian of system. What can you say about total angular momentum?

(xi)- A hydrogen atom can be viewed as two particle system – a proton and an electron with Coulomb interacting potential between them. Write the Schrodinger wave equation for such a system by explaining all the terms used.

(xii)- Radial distribution function is given by,

$$P(r) = \frac{4}{a_0^3} r^2 e^{-2r/a_0}$$

Find the most probable distance.

(xiii)- Why is a non-uniform magnetic field used in the Stern–Gerlach experiment?

(xiv)- For Pauli spin matrix $\sigma_x$, prove the relation,

$$e^{i\theta\sigma_x} = \hat{I}\cos\theta + i\theta_x \sin\theta$$

(xv)- Write Pauli spin matrices without derivation. Also give single important relation obeyed by them.

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**Q.2. Solve the following. (3x10=30)**

(i)- State and prove generalized uncertainty relation (Uncertainty principle for operators).

(ii)- Express angular momentum in spherical polar coordinates and show that,

$$\hat{L}^2 = -\hbar^2\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right\}$$

(iii)- Define the term central potential. Starting with time independent Schrodinger wave equation, obtain radial wave equation for central potentials.

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